

3 The nervous system coordinates and regulates body functions.

Fig. 3.1 shows two diagrams, each showing parts of the nervous system in a human.

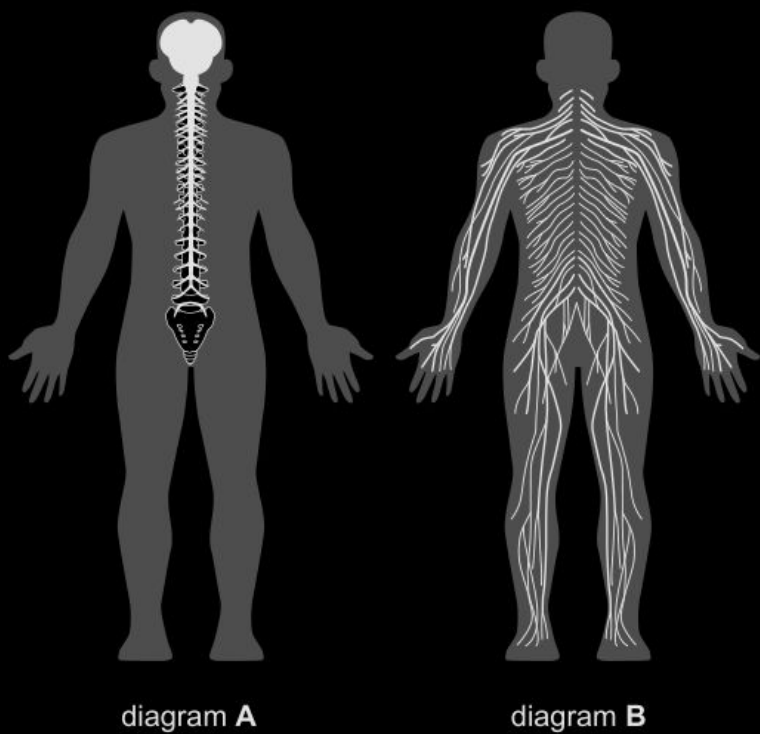


Fig. 3.1

(a) Describe the parts of the nervous system shown in each diagram.

diagram A

.....

diagram B

.....

[4]

(b) Fig. 3.2 shows the junction between two neurones.



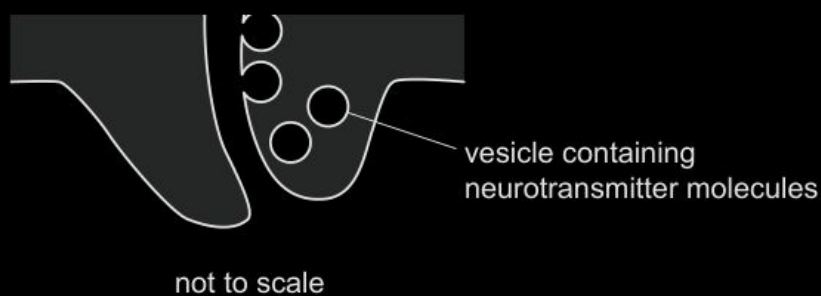


Fig. 3.2

- (i) Name the junction shown in Fig. 3.2.

..... [1]

- (ii) Draw an arrow on Fig. 3.2 to show the direction that an impulse will travel. [1]

- (iii) The distance between the two neurones shown in Fig. 3.2 is 0.00003 mm.

State this distance in units of μm .

Space for working.

..... μm [1]

- (c) Nicotine is a chemical found in cigarette smoke.

A molecule of nicotine has a similar shape to a molecule of neurotransmitter.

Suggest and explain the effect of nicotine at the junctions between neurones.

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..... [3]

[Total: 10]

3(a)	(<i>diagram A</i>) central nervous system / CNS ; <u>brain + spinal cord</u> ; (<i>diagram B</i>) peripheral nervous system / PNS ; nerves / nervous tissue / neurones + outside CNS AW ;	4	
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3(b)(i)	synapse ;	1	Allow synaptic cleft / synaptic gap
3(b)(ii)	 ;	1	
3(b)(iii)	0.03 ;	1	
3(c)	1 nicotine + binds / blocks AW ; 2 receptor (on post-synaptic membrane / relay) ; 3 protein ; 4, <u>impulse</u> + generated / produced / travel / more AW ;	3	

Question	Answer	Marks	Guidance
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3 Fig. 3.1 shows the central nervous system and some other parts of a man's body.

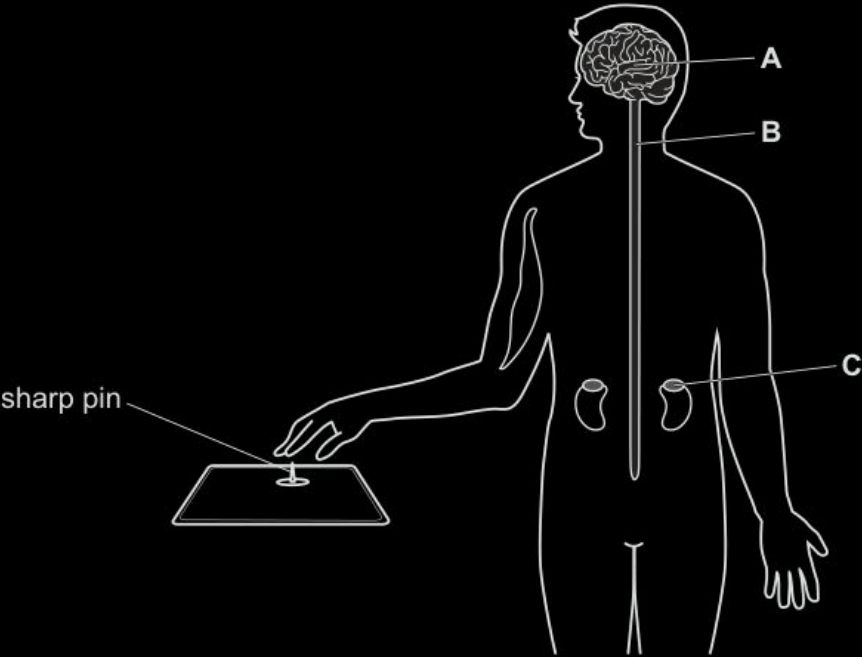


Fig. 3.1

(a) Name the **two** parts of the central nervous system labelled **A** and **B**.

A

B

[2]

(b) The man accidentally touches his hand on a sharp pin and immediately lifts his lower arm away from the pin in a reflex action.

(i) This reflex action involves a receptor and an effector.
On Fig. 3.1, draw a label line to the location of:

- the receptor and label it **R**
- the effector and label it **E**.

[2]

(ii) Describe the roles of **two** named neurones in this reflex arc.

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[4]

(c) The part of the body labelled **C** is a gland in the endocrine system.

(i) Name gland **C**.

..... [1]

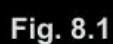
(ii) Explain the role of gland **C** in triggering an increase in heart rate in response to a sudden

3(a)	A = <u>brain</u> ; B = <u>spinal cord</u> ;	2	
3(b)(i)	R pointing to the hand ; E pointing to muscle ;	2	

3(b)(ii)	<i>sensory neurone</i> <i>idea that</i> a sensory neurone receives impulses / is stimulated by the receptor ; carries impulses to spinal cord / CNS ; <i>relay neurone</i> carries impulses from sensory neurone ; to motor neurone ; <i>motor neurone</i> carries impulses from relay neurone / spinal cord / CNS ; to the effector / muscle ;	4	no neurones named = no marks if more than two neurones described, mark best two named neurones
3(c)(i)	adrenal (gland) ;	1	
3(c)(ii)	any two from: produces a hormone / adrenaline ; travels in blood ; prepares the body for action / prepares body for increased activity ;	2	

Question	Answer	Marks	Guidance
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The sensory neurone receives information from a pain receptor.



[6]

(b) Analgesics are drugs that help to stop a person from feeling pain.

Fig. 8.2 shows the shape of some analgesic drug molecules.





Fig. 8.2

Suggest how this analgesic drug can help to stop a person from feeling pain.

Use information from Fig. 8.1 and Fig. 8.2 in your answer.

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..... [4]

[Total: 10]

8(a)	max six from: use of neurotransmitter molecules ; impulse stimulates the release of neurotransmitter molecules ; from vesicles ; into the synaptic gap / cleft ; the neurotransmitter molecules pass across the gap / cleft ; by diffusion ; then bind with receptor (proteins / sites) ; cause an impulse to be initiated in the relay / post synaptic neurone ;	6	
8(b)	the analgesic / drug has a similar shape to the neurotransmitter ; fits / binds with the receptor (proteins / sites) ; this blocks the receptor (protein / sites) / stops the neurotransmitter binding ; information from the pain receptor cannot reach the brain ;	4	

9 (a) Describe the similarities and differences in the **functions** of a motor and a sensory neurone.

similarities

.....

.....

differences

.....

.....

[5]

9(a)	<i>similarities:</i> both carry impulses ; quickly ; over long distances ; during voluntary / involuntary / reflex actions ; <i>differences :</i> <i>motor</i> – from CNS / spinal cord / brain ; to effector / named effector ; <i>sensory</i> – from receptor / named receptor ; to CNS / spinal cord / brain ;	5
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9 (a) For each of the following, state where, in the reflex arc, they are found and their functions.

- sensory neurones

.....

.....

- relay neurones

.....

.....

- motor neurones

.....

.....[5]

(b) Explain why, just after hearing a sudden loud noise, a person's heart beats faster.

.....

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.....

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.....

.....

.....[5]

[Total: 10]

- 9 (a) impulses anywhere;
Accept specific examples for next 6 mps
(sensory) from sense organ/receptor;
to CNS/spinal cord/brain;
(relay) within CNS;
connecting sensory to motor neurones;
(motor) from CNS/spinal cord/brain;
to effector/muscle/gland;

[max 5]

- (b) sound + stimulus;
ears AW;
one correct reference to a neurone;
brain;
gland/adrenal/suprarenal;
adrenaline/epinephrine;
blood;
heart muscles;
ref. fight/flight/fright etc. response – or described;

[max 5]

[Total: 10]

(a) (i) Name the type of coordinated response that takes place as a result of a specific stimulus, such as the withdrawal of the hand from a hot object touching the skin.

..... [1]

(ii) Outline, in the correct sequence, the functions of each **named** type of neurone involved in producing this coordinated response.

(b) Suggest why damage to a person's spinal cord may result in the person not being able to produce coordinated responses.

[Total: 10]

7(a)(i)	reflex / involuntary ;	1	
7(a)(ii)	1 <u>impulse</u> ; 2 <u>synapse</u> ; 3 from + <u>receptor</u> ; 4 sensory / afferent ; 5 relay / inter / intermediate ; 6 CNS / spinal cord ; 7 motor / efferent ; 8 to + <u>effector</u> ; 9 muscle / named muscle correct for withdrawal reflex ; 10 sequence mark: point(s) 3/4/5/6 <u>must be ticked before</u> point(s) 7/8/9 ;	7	
7(b)	<u>impulse</u> + not transmitted AW ; effector / named effector + not activated AW ;	2	

Question	Answer	Marks	Guidance
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(b) Describe how **named** components of the nervous system are involved in producing a reflex action.

[Total: 10]

8(a)	named stimulus / trigger for a specific reflex action ; correct named receptor for stimulus given ; action described correct for example given ; importance of specific action explained ;	4	
8(b)	synapse (anywhere in sequence) ; impulse / electrical pulse (anywhere in sequence) ; receptor ; detection of stimulus ; then sensory neurone ; relay / inter(mediate) / connector neurone ; reference to CNS / brain / spinal cord ; then motor neurone ; effector / named effector ; action of effector or described ;	6	1g signal / message

Question	Answer	Marks	Guidance
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3 The following is a list of terms associated with a person's responses. Each term is to be used only once. Write the letter of the term next to the response. Each response is worth 2 marks.

- | | |
|-----------------|---------------------|
| A – brain | F – motor neurone |
| B – contraction | G – muscle |
| C – gland | H – receptor |
| D – hormone | I – sensory neurone |
| E – impulse | J – spinal cord |

(a) Using their identifying letters only, place the terms in the order in which they are involved in the following:

(i) Peeling an onion causes a student's eyes to water.
.....[2]

(ii) A student decides to open a book.
.....[1]

(iii) A student hears a sudden loud noise and, shortly afterwards, his heart beats faster.
.....
.....[2]

(b) Name the type of response involved in (a)(i) and (a)(ii).

(a)(i)
(a)(ii)[2]

(c) Describe how responses involving the nervous system differ from those that involve the effects of hormones.

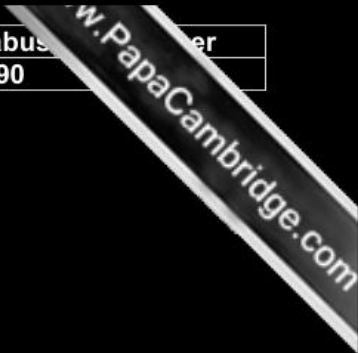
.....
.....
.....
.....[3]

[Total: 10]



- 3 (a) (i) **H, E, I** (or **I, E**), **A, F, C** all correct = 2, 3 in correct consecutive sequence =1 (Ignore any other 'E's'); [2]
- (ii) **A, E, J, F, G, B** (Ignore any other E's); [1]
- (iii) **H, E, I** (or **I, E**), **A, F, C, D, G** (**B**) (4 in correct consecutive sequence = 1) (Ignore any other 'E's' (which can include the final **B**)); [2]

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- (b) reflex / involuntary / unintentional (R *spinal* reflex);
voluntary / deliberate / intentional;
- (c) faster / fast v. slow;
electrical / impulses v. chemical / blood;
neurones v. blood / chemical;
(R nerves)
muscles v. glands / organs;
temporary AW v. more lasting AW;
sometimes deliberate AW v. less controlled AW;
specific organs v more general effect;
- [max 3]

[Total: 10]

2 Fig. 2.1 shows a section through the human eye.

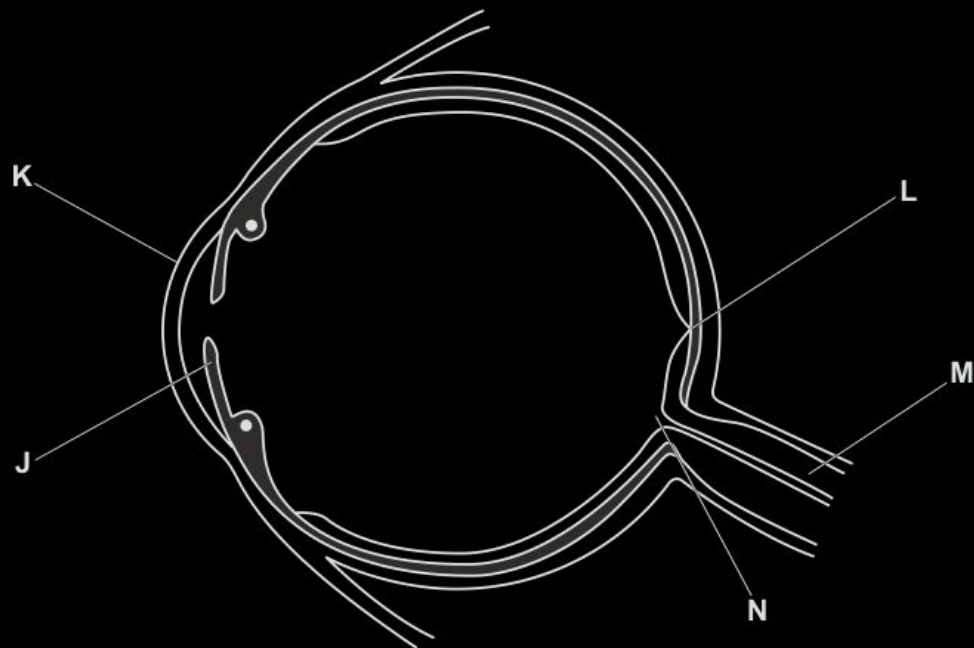


Fig. 2.1

- (a) (i) Identify the letter on Fig. 2.1 that has been used to label the part of the eye that:
- contains circular and radial muscles
- has the highest concentration of light receptors.
- [2]
- (ii) On Fig. 2.1, draw **and** label:
- the lens
 - the suspensory ligaments.
- [2]
- (iii) Explain the function of the suspensory ligaments.
-
-
-
-
-
-
-
- [3]

2(a)(i)	J ; L ;	2	
2(a)(ii)	lens labelled and drawn in correct position ; suspensory ligaments labelled and drawn in correct position ;	2	
2(a)(iii)	max three from: attached to ciliary bodies / muscles ; can be loose / slack or taut / tight ; pulls on the lens / ora ; controls / changes the shape of the lens ; allows <u>accommodation</u> to occur ;	3	

2(b)(i)	the light will not be bent / refracted enough ; light will not be focussed on the retina ;	2	
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4 Fig. 4.1 shows changes in the curved shape of the lens of a person's eye over time.

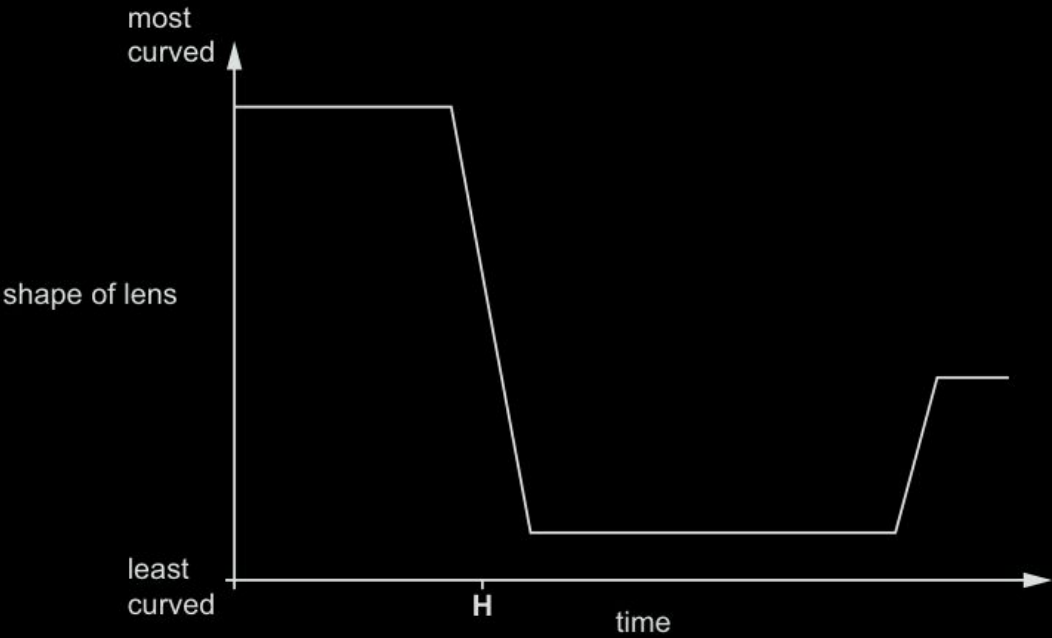


Fig. 4.1

- (a) (i) During the period of time shown in Fig. 4.1, the person carried out each of the following actions:
- P read the words in a book
 - Q read the time on a wall clock
 - R looked out of a window at a bird high in the sky.

Identify in what order the person carried out actions **P**, **Q** and **R**.

Place the letters **P**, **Q** and **R** in the correct order below.

..... → → [1]

- (ii) At the time labelled **H** on Fig. 4.1, the shape of the lens is changing. Outline, with reference to **named** structures in the eye, how this change happens.

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..... [4]

- (b) Fig. 4.2 shows a front view of the eye in dim light.





Fig. 4.2

- (i) Complete the labels on Fig. 4.2 by naming the two parts of the eye indicated by the label lines. [2]
- (ii) Complete Fig. 4.3 to show a view of the same eye as it would appear in bright light. [1]

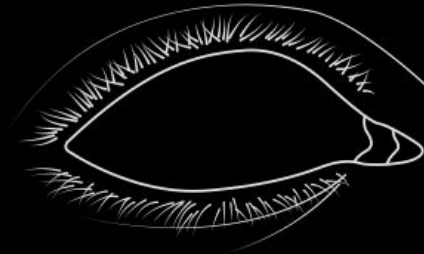


Fig. 4.3

- (iii) State the type of nervous response that takes place when the eye changes its appearance from that in Fig. 4.2 to that in Fig. 4.3.

..... [1]

- (iv) Suggest the importance to the person of this response taking place in the eye.

.....

.....

..... [2]

[Total: 11]

4(a)(i)	P – R – Q ;	1	
4(a)(ii)	<u>ciliary muscles</u> ; (ciliary muscles) <u>relax</u> ; <u>suspensory ligaments</u> ; (suspensory ligaments) tighten / under tension / taut ;	4	

4(b)(i)	pupil ; iris ;	2	
4(b)(ii)	pupil drawn smaller diameter ;	1	
4(b)(iii)	<u>reflex</u> ;	1	
4(b)(iv)	any two from: protect ; rapidly ; retina / rod cells / cone cells / light receptors + from too much light / high light intensity / bright light ;	2	

Question	Answer	Marks	Guidance
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(c) When the bright light is shone into the eye there is a pupil reflex which causes the pupil to become smaller.

Explain how named parts of the eye and of the nervous system are involved in this pupil reflex.

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.....

..... [4]

2(c)	<i>any four from:</i> retina / rods detect extra light (intensity) ; impulse + to CNS / brain ; via optic nerve / sensory neurone ; back to eye / iris / effector / muscle + motor neurone ; radial muscles relax ; circular muscles contract ; iris becomes bigger / elongates ; reflex arc ;	4	
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4

Fig. 4.1 shows a horizontal section of the human eye and the pathway taken by light rays as they leave an object.

- (a) Complete the diagram by continuing the lines from the object to show how the light rays produce a focussed image on the retina.

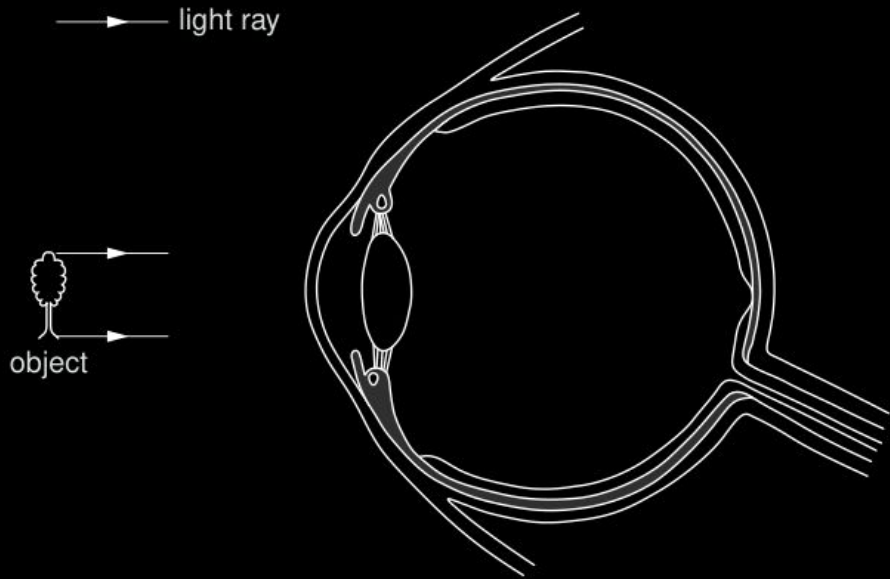


Fig. 4.1

[3]

- (b) (i) State how the appearance of the pupil in the eye will change when a person moves from dim light into an area of bright light.

..... [1]

- (ii) Explain how this change is brought about by structures in the eye.

.....
.....
..... [2]

- (c) The change in appearance of the pupil when entering an area of bright light is a reflex action.

- (i) Define the term *reflex action*.

.....
..... [2]

- (ii) Suggest why drugs that prevent this reflex action from occurring should be avoided.

.....
.....
..... [2]





- 4 (a) rays continue parallel until hit cornea;
rays converge at cornea;
rays converge at lens;
meet before retina + continue to hit retina;
- (b) (i) narrows / decreases in size or diameter / constricts (R contracts); [1]
(ii) iris + muscles;
circular + contract; [2]
- (c) (i) fast / rapid / quick;
reaction / response + (to) stimulus;
automatic / involuntary / no involvement of conscious thought AW; / cannot be controlled
R no involvement of brain [2]
- (ii) too much light allowed to enter AW;
damage to retina / rods / cones / light-sensitive cells; [2]

[Total: 10]

1 Fig. 1.1 shows the human eye in horizontal section.

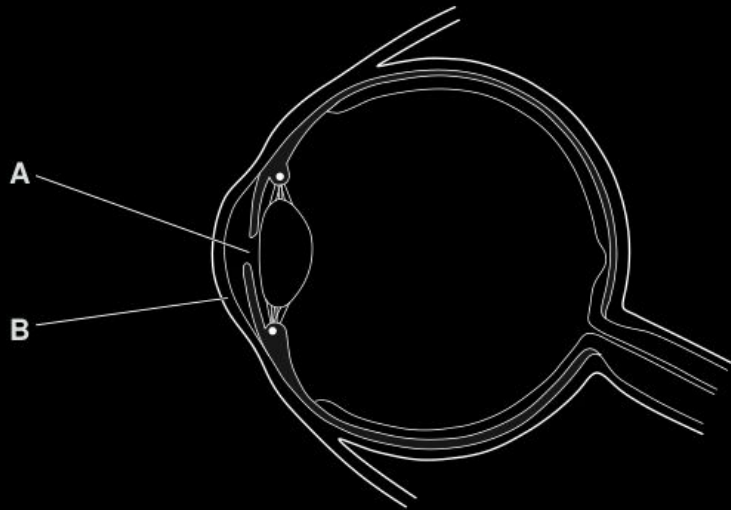


Fig. 1.1

(a) (i) Identify **A** and **B** that are labelled on Fig. 1.1.

A

B [2]

(ii) Describe what happens to **A** when light entering the eye becomes less intense.

.....
..... [1]

(iii) Place a letter **Z** on Fig. 1.1 where a response occurs as a result of a reflex action. [1]

(b) In some people's eyes, the retina becomes completely detached from the tissues beneath. Explain how this will affect their ability to see.

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.....
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[3]

(c) As people get older, cloudy (opaque) patches sometimes form in the lens of the eye. These are called cataracts. Suggest how cataracts might affect the ability of the lens to carry out its function.

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.....
.....
..... [3]

[Total: 10]





- 1 (a) (i) A pupil / aqueous humour / anterior chamber;
B cornea;
- (ii) dilates / gets big / enlarges / opens / gets wide / expands;
- (iii) Z on ciliary muscle / iris / external muscle;

[1]
[1]

- (b) reasonable ref. light receptors / sensitive cells / rods / cones;
not in contact with optic nerve;
no impulses;
to brain;
no picture formed / unable to see / blindness;
(R blurred vision or reduced visionary powers)

less / no nutrition for retina; [max. 3]

- (c) failure to focus (all) light (rays);
blurred image AW;
any reference to the passage of light rays being impaired
(e.g. reflection / refraction / deflection / absorption / convergence);
ref. possible change in elasticity / ability to accommodate;
faded colour vision;

[max. 3]

- (b) A man is threading a needle.



To do this, each of the man's eyes produces a focused image of the needle.

Explain how different parts of the eye function to produce a focused image of a near object.

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..... [5]

7(b)	light is refracted / bent by cornea ; ciliary muscles / body contract ; suspensory ligaments become slack ; lens becomes fatter AW ; light refracted / bent (more) by lens ; light / image focused / formed on retina ; light detected by light receptors in retina / rods / cones ; nerve impulses triggered ;	5	
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1 (a) Hormones are produced by glands in the body. They alter the activity of specific target organs.

Draw lines to link:

- each gland to a hormone it produces
- each hormone to a specific target organ.

Two lines have been drawn for you. Draw **six** more lines.

gland	hormone	target organ
adrenal	glucagon	ovary
ovary	progesterone	uterus
pancreas	follicle-stimulating hormone (FSH)	heart
pituitary	adrenaline	liver

[5]

1(a)	adrenal glucagon ovary ovary progesterone uterus pancreas FSH heart pituitary adrenaline liver	5	left side of diagram: number of marks = number of correct lines right side of diagram: 2 correct lines = 1 mark ----- = given to candidates in the question any additional lines to or from a box negates that mark.
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1 The human body produces over fifty different hormones.

(a) Describe what is meant by the term hormone.

.....

.....

.....

..... [3]

(b) The list shows some of the hormones produced by the human body:

- A adrenaline
- B luteinising hormone
- C insulin
- D progesterone
- E oestrogen
- F glucagon.

Complete Table 1.1 by writing the correct letters in each box. Each box can contain one or more letters and each letter can be used more than once. The first one has been done for you.

Table 1.1

	hormones
produced in the pancreas	C and F
produced in the pituitary gland	
produced in the ovary	
released to increase blood sugar concentration	
released to build up the uterus lining after menstruation	

[4]

[Total: 7]

1(a)	a chemical / substance produced / released by a gland ;	1	
	carried by blood ;	1	
	alters activity of an organ ;	1	A affects target organs
1(b)	B ;	1	
	D and E ;	1	
	A and F ;	1	
	E ;	1	Ig D

Question	Answer	Marks	Guidance
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5090/21/M/J/22 Q8

8 The functions of the body are coordinated by nerve impulses and hormones.

(a) Compare these two types of coordination.

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..... [4]

(b) A girl sees a vehicle travelling towards her at speed as she crosses a road. This triggers nerve impulses and the release of adrenaline in her body.

Describe the role of the nerve impulses and adrenaline in helping her to react to this situation.

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..... [6]

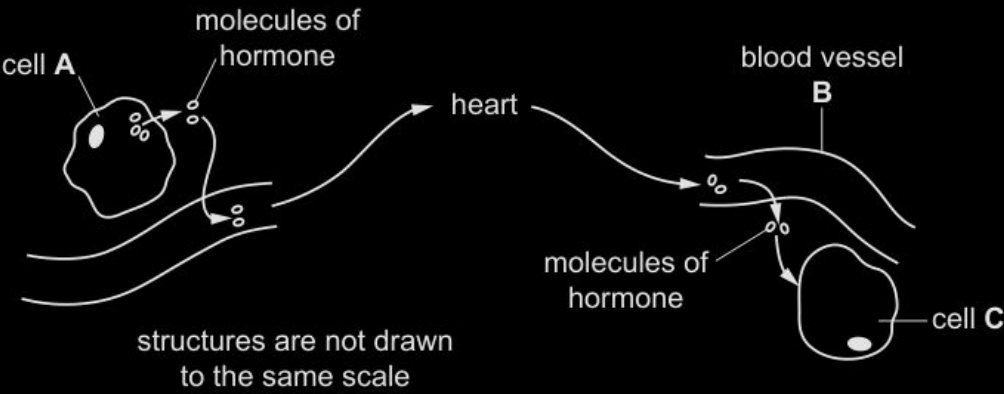
[Total: 10]

8(a)	any four from: 1 both trigger responses ; 2 both work on specific effectors AW ; 3 nerve impulses fast + hormones slow ; 4 nerve impulses electrical + hormones chemical ; 5 nerve impulses short term effect + hormones longer term effect ; 6 nerve impulses travel along nerve cells / neurones AW OR hormones travel in blood / plasma ; 7 nerve impulses all the same / hormones are different chemicals ; 8 nerve impulses generated by receptors AW / hormones from glands ;	4	} comparative statements required accept faster / slower etc.
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8(b)	any six from: eye <u>receptors</u> / retina / rods / cones ; impulse + sensory neurones ; via relay neurones ; to motor neurones ; to effector / muscle ; adrenaline + from adrenal glands ; travels in blood + to named specific target organ e.g. heart / lungs ; heart beats faster ; stimulates glycogen to be converted to glucose ; more glucose / oxygen / blood + to muscles ; air passageways increase in diameter ; blood diverted AW + to muscles ; pupils dilate ;	6	
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Question	Answer	Marks	Guidance
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1 The diagram shows the pathway taken by a hormone from cell A where it is produced to cell C that it affects.



(a) (i) Identify the type of blood vessel B. [1]

(ii) Explain the ways in which this type of blood vessel is adapted for the transport of hormone molecules.

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.....
.....
.....
..... [4]

(iii) State how molecules of the hormone are transported through blood vessel B.

.....
..... [2]

(b) For a **named** hormone, identify the gland containing cell A, an organ containing cell C and a function or effect of the hormone you have named.

- named hormone
- the gland containing cell A
- an organ containing cell C
- a function or effect of the named hormone

.....
.....

(c) Cell C is one of a number of similar cells making up an organ.

State the name given to organs affected by hormones.

..... [1]

[Total: 12]

MS 5090/21/M/J/20 Q1

1(a)(i)	capillary ;	1
1(a)(ii)	very small AW / narrow ; thin / one cell thick / leaky + walls ; runs close to the cells ; very numerous / large surface area ; maximum uptake of hormone / maximum offloading of hormone AW ; by diffusion ;	4
1(a)(iii)	in solution ; in blood ; plasma ;	2
1(b)	(for example) insulin ; pancreas ; liver / muscles ; glucose to glycogen / glycogen storage ;	4
1(c)	target (organ) ;	1
Question	Answer	Marks

3 (a) Use words from the list to complete the paragraph.

- platelets glucagon pancreas plasma
- liver sucrose endocrine
- pituitary insulin glycogen

Glucose is transported in solution by the component of blood called If the concentration of glucose in the blood increases above normal, the hormone is released into the blood by an gland called the This hormone promotes the removal of glucose from the blood and its conversion into the chemical

[5]

3(a)	plasma ; insulin ; endocrine ; pancreas ; glycogen ;	5	
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3 A man is cooking in a hot kitchen and sweat is being released by sweat glands in his skin.

(a) (i) Explain the term homeostasis with reference to sweating.

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..... [2]

(ii) Explain how receptors and blood vessels in the man’s skin are also involved in homeostasis when he is in the hot kitchen.

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.....

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..... [4]

(b) Sweat contains very small quantities of the waste product urea.

State the name of the organ where most urea is made in the body.

..... [1]

(c) The table shows the concentrations of some chemicals found in the sweat, urine and blood plasma of a healthy human.

chemical	concentration of chemical /mmol per dm ³		
	sweat	urine	blood plasma

urea	22	393	6
sodium	66	110	141
chloride	59	103	99

- (i) Use the data in the table to compare the chemical composition of urine and blood plasma.

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..... [4]

- (ii) A patient has kidney disease and is about to start dialysis treatment.

Suggest and explain the concentration of urea in the sweat of the patient compared to that of a person who does not have kidney disease.

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.....

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..... [3]

[Total: 14]

3(a)(i)	<i>any two from:</i> (homeostasis is) the maintenance of a <u>constant</u> ; <u>internal environment</u> / temperature ; return to norm / return to set-point / negative feedback ; <u>evaporation</u> of sweat removes heat / thermal energy / cools / decreases temperature ;	2	
3(a)(ii)	<i>any four from:</i> receptors + detection of temperature ; <u>impulse</u> to brain / CNS ; <u>vasodilation</u> ; arterioles / blood vessels widen ; increase in blood + near skin surface ; greater + loss of heat / thermal energy ; by radiation ;	4	
3(b)	<u>liver</u> ;	1	

3(c)(i)	<i>any four from:</i> less urea in blood plasma ; more sodium in blood plasma ; less chloride in blood plasma ; comparative data quote with units ; data manipulation, e.g. 50 × higher urea in urine than blood plasma ;	4	
3(c)(ii)	high(er) / over 22 (in patient) ; reduced filtration / reduced removal of urea (from kidney) ; reference to blood <u>plasma</u> ;	3	

Question	Answer	Marks	Guidance
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2 Fig. 2.1 shows changes in the body temperature of a person.

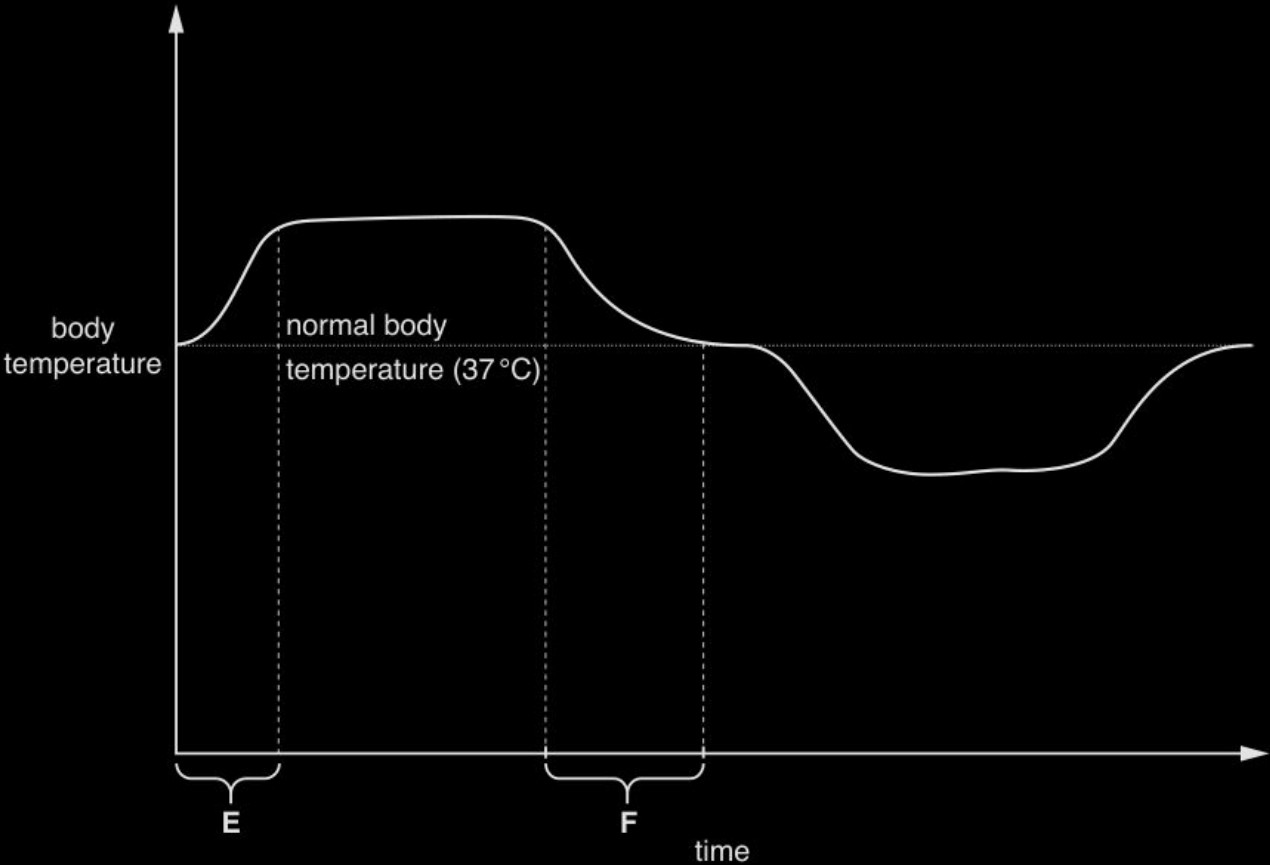


Fig. 2.1

- (a) State the term for maintaining constant conditions, such as temperature, in the body.
.....[1]
- (b) Suggest **three** things that could happen to account for the shape of the curve during time **E**.
- 1
- 2

2(a)	homeostasis ;	1	
2(b)	1 body temperature increased ; 2 exercise AW ; 3 increased / high + surrounding AW temperature ; 4 increased / high + surrounding AW humidity ; 5 hot + food / drink ; 6 increased / high + metabolism / respiration ; 7 fever / illness ; 8 ovulation ; 9 wear more clothes AW ;	3	
2(c)	1 sweat ; 2 evaporation ; 3 vasodilation AW ; 4 reduced + metabolism / respiration ; 5 more blood + to skin ; 6 heat loss / radiation ;	3	
2(d)(i)	negative feedback ;	1	
2(d)(ii)	hypothalamus ;	1	

Question	Answer	Marks	Guidance
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6 (a) Define the term *homeostasis*.

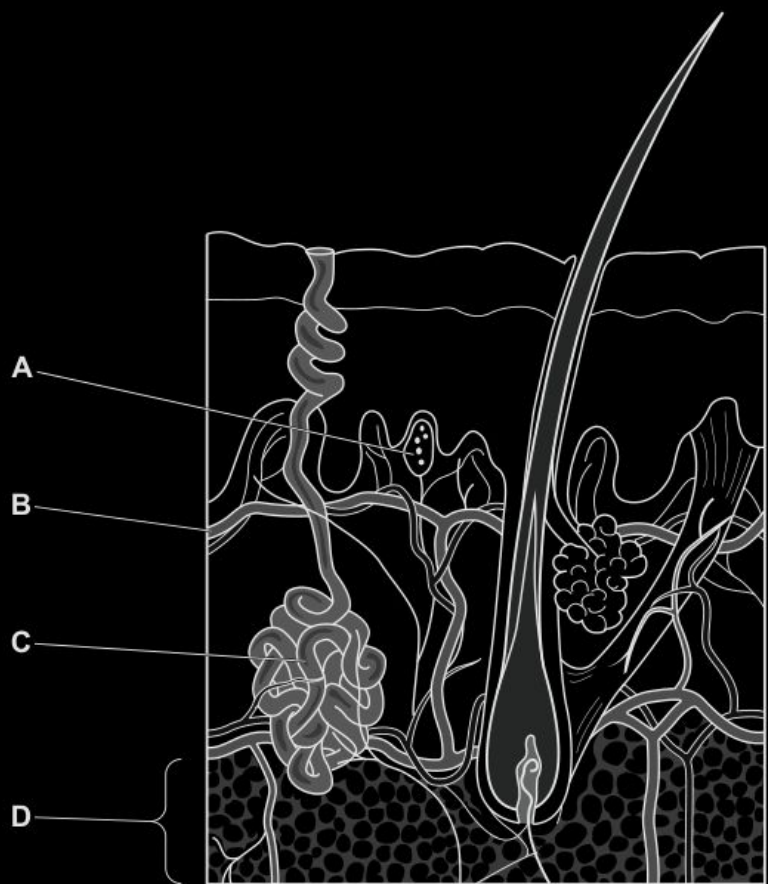
(b) Use the concept of control by negative feedback to explain what would happen in each of the following situations.

(i) A person goes outside wearing only short trousers and a short-sleeved shirt on a day when the air temperature is below 10°C.

6	(a)	maintenance of / constant ; internal environment / conditions within the body ;	[2]	A regulating / control lg specific examples
	(b) (i)	low temperature detected ; by receptors / sensors on skin ; brain ; <i>any two corrective mechanisms from</i> shivering / hairs raised / sweating reduced / stopped / vasoconstriction of blood vessels (R of capillaries) ;; temperature rises again ;	[4]	A hypothalamus



1 The diagram shows human skin.



(a) (i) Identify the parts labelled **B** and **C**.

B

C

[2]

(ii) Describe how part **B** is involved in maintaining a constant body temperature when the external temperature drops.

.....
.....
.....
.....
..... [3]

(b) Part **A** is unevenly distributed in human skin.

Name part **A** and suggest reasons for this uneven distribution.

.....
.....
.....
..... [3]

(c) The thickness of layer **D** can change over several months.

Suggest and explain **one** reason for a **decrease** in the thickness.

.....

.....

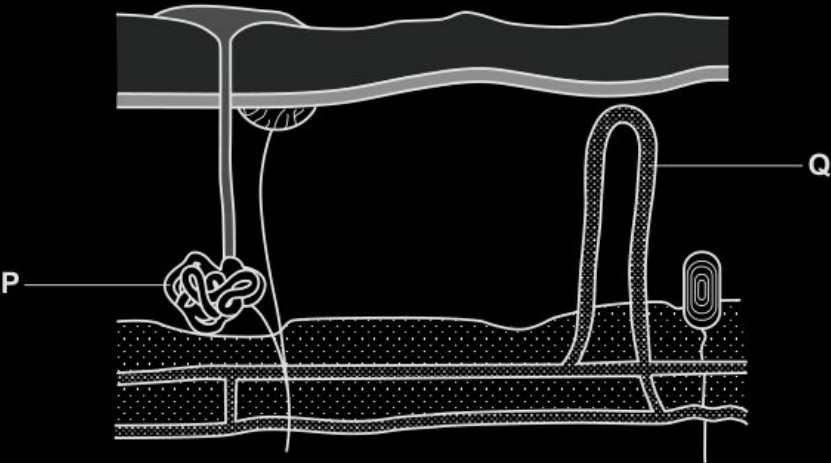
..... [2]

[Total: 10]

1(a)(i)	B = blood vessel / capillary / artery / arteriole ;	1	
	C = sweat gland ;	1	
1(a)(ii)	less blood flows + in blood vessel / close to skin surface ;	1	
	decrease in radiation / heat loss / AW ;	1	
	vasoconstriction OR blood vessels / arterioles / arteries / part B + narrow / constrict ;	1	
1(b)	<i>any three from:</i> A is a receptor / nerve ending ; touch / pressure / pain / temperature ; examples of uneven distribution e.g. high density in fingers / lips ; exposed areas + need higher sensitivity AW ;	3	
1(c)	<i>any two from:</i> fatty / adipose + tissue / cell ; less food eaten / less fat in diet / exercise; so fat used by body + for respiration / release energy ; body using more energy than it is taking in ;	2	

Question	Answer	Marks	Guidance
----------	--------	-------	----------

6 The diagram shows a vertical section through human skin in a hot environment.



(a) Name **and** outline the role of **P** and **Q** in the control of body temperature when a person is in a hot environment.

P

.....

.....

.....

.....

.....

Q

.....

.....

.....

.....

.....

[7]

(b) Explain how the process of temperature control in humans is an example of:

homeostasis

.....

negative feedback.

.....

.....

[3]

MS 5090/22/O/N/20 Q6

6(a)	(allow once in either section P or section Q) 1 removal / loss of + heat ; (allow anywhere in section P) 2 sweat gland ; 3 produce / release + sweat ; 4 evaporates / vapour ; (allow anywhere in section Q) 5 capillary ; 6 increased AW + blood flow ; 7 blood near + surface AW ;	7	
6(b)	(homeostasis) 1 constant / same + internal environment AW / temperature ; (negative feedback) 2 change / increase / decrease ; 3 correction / reversal / back (e.g. down / up / to normal) AW ;	3	

Question	Answer	Marks	Guidance
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1

Fig. 1.1 shows a vertical section through the skin in two different environmental conditions, **A** and **B**.

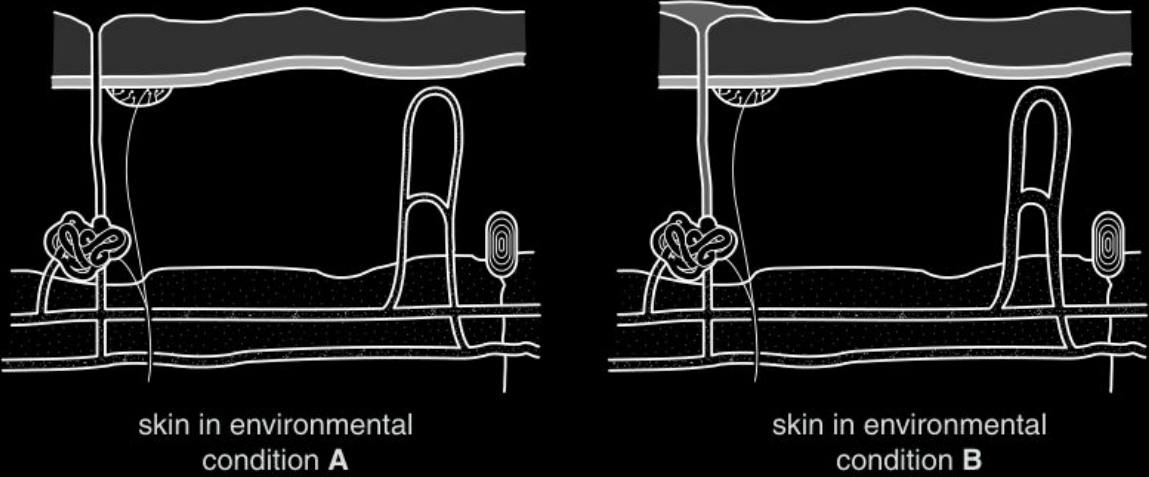


Fig. 1.1

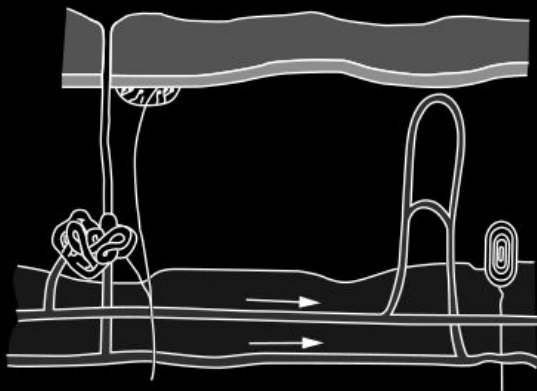
- (a) On Fig. 1.1, label each of the following:
- a sweat gland
 - a capillary. [2]
- (b) Use the information in Fig. 1.1 to suggest how environmental condition **B** is different from environmental condition **A**.
-
- [1]
- (c) (i) State **two** differences between the skin in environmental condition **A** and the skin in environmental condition **B**.
- 1
-
- 2
- [2]
- (ii) Explain the advantages to a person in environmental condition **B** of the **two** differences you have stated in your answer to (c)(i).
-
-
-
-
-
-

..... [3]

[Total: 8]

1	(a) (i)	sweat gland labelled ; capillary labelled ;	R sweat duct	[2]
	(b)	warmer ;	must be a comparative statement	[1]
	(c) (i)	<u>sweat</u> (present in B / ORA) ; capillaries carry more blood (in B / ORA) ; capillaries wider / dilated (in B / ORA) ;	lg constrict / shrink	[max 2]
	(ii)	evaporation (of sweat) ; more blood near (surface of) skin ; increased heat loss / cooling ; allows regulation of temperature / prevents overheating ;		[max 3]
				[Total 8]

1 Fig. 1.1(a) and Fig. 1.1(b) show a section through the skin of a person at two different body temperatures.



Key

→
direction of
blood flow

Fig. 1.1(a)

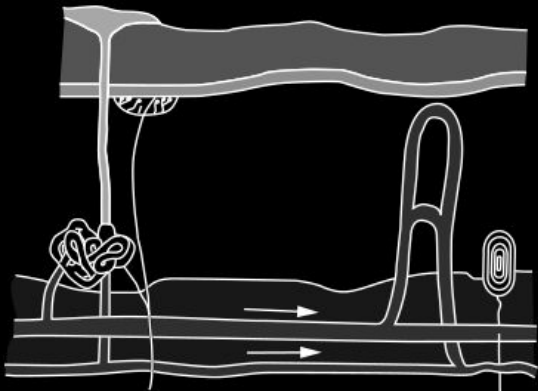


Fig. 1.1(b)

- (a) On Fig. 1.1(a), identify and label each of the following.
- (i) a sweat gland
 - (ii) a temperature receptor. [2]
- (b) (i) State what has happened to the body temperature to cause the changes shown in Fig. 1.1(b) compared with Fig. 1.1(a). [1]
-
- (ii) State **two** reasons for the answer you have given in (b)(i).
- 1
- 2 [2]
- (c) (i) On Fig. 1.1(a), label and name the type of blood vessel that constricts when body temperature falls.
- (ii) Explain the effects caused by constriction of the blood vessel you have named in (c)(i). [4]
-
-
-
-

1 (a) (i)	label line must touch the sweat gland;	[1]	A label on gland or duct A (i) = gland
(ii)	label line must touch some part of receptor under Malpighian layer;	[1]	(ii) = temp receptor
(b) (i)	(temperature / it) increases / rises;	[1]	R cooler / decreases OWTTE
(ii)	In either order: sweat; wider blood vessels / capillaries / more blood;	[2]	R veins
(c) (i)	arteriole; the upper horizontal blood vessel labelled;	[2]	A Artery (the word – ignore what is labelled) (ignore name given to it)
(ii)	less blood (flows); to capillaries (R capillaries constrict) to sweat glands; blood carries heat / less heat carried <u>less</u> heat lost by skin / radiation ; <u>less</u> sweat produced; ref to latent heat of vaporisation;	[max 4]	A less heat lost by blood
		[Total 11]	

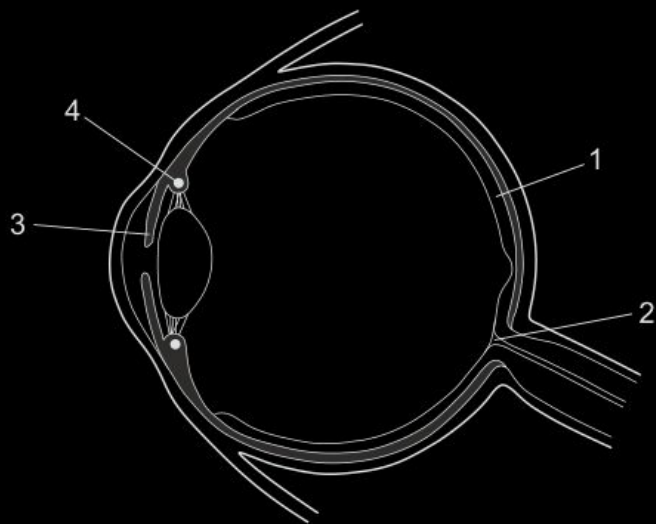
Page 4	Mark Scheme	Syllabus	Paper
	GCE O LEVEL – October/November 2013	5090	21

(b) Describe the role of the hormone adrenaline and give **one** example of a situation in which adrenaline may be released.

[Total: 10]

9(a)	hypothalamus ; reference to detecting blood temperature ; receptors ; in skin ; detect temperature + of surroundings AW ; impulse ; sensory neurone ; to central nervous system / brain / spinal cord ; motor neurone ; <u>effector</u> ; named effector ; shiver / vasoconstriction or vasodilation / sweat increase or decrease ; reference to negative feedback ;	6	A muscle or named effector / sweat gland
9(b)	correct named situation ; target organ / named ; (any two effects from) glycogen to glucose / sugar ; increased (heart) rate ; pupils dilate ; dry mouth ; breathing rate change ; diversion of blood to muscle / from gut ;	4	increased heart rate = 2 marks (named target organ + effect)

24 The diagram shows a section through the eye.



In the pupil reflex, which row gives the sites of the effectors and receptors involved?

	effectors	receptors
A	3	1
B	3	2
C	4	1
D	4	2

24	A	1
----	---	---

26 Which process is stimulated by adrenaline in the cells of the liver?

- A breakdown of glycogen, increasing the blood glucose level
- B breakdown of excess amino acids, forming urea
- C breakdown of proteins, releasing amino acids into the blood
- D conversion of excess blood glucose to glycogen

MS 5090/11/M/J/24 Q26

26	A	1
----	---	---

5090/12/M/J/24 Q23

23 What is a function of sensory neurones?

- A transmitting impulses from muscles to the spinal cord
- B transmitting impulses from receptors to muscles
- C transmitting impulses from receptors to the spinal cord
- D transmitting impulses from the spinal cord to muscles

MS 5090/12/M/J/24 Q23

23	C	1
----	---	---

24 Descriptions of changes that occur in the eye are listed.

- 1 Ciliary muscles contract.
- 2 Ciliary muscles relax.
- 3 Suspensory ligaments become slack.
- 4 Suspensory ligaments tighten.
- 5 The lens becomes more spherical.
- 6 The lens becomes thinner.
- 7 Light rays are refracted less.
- 8 Light rays are refracted more.

Which row lists the changes that occur when focusing on distant and near objects?

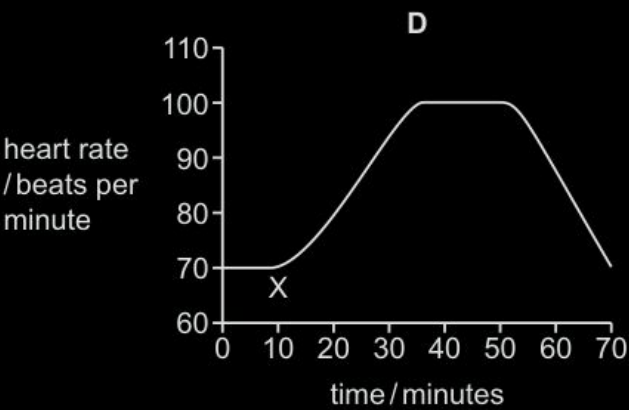
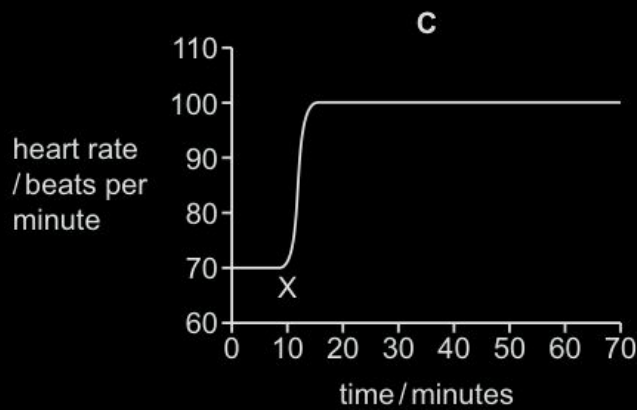
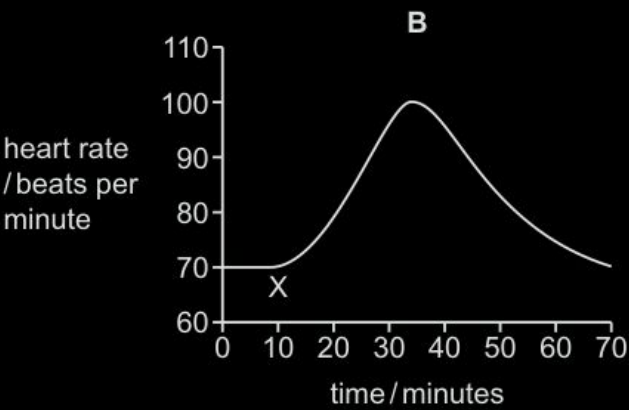
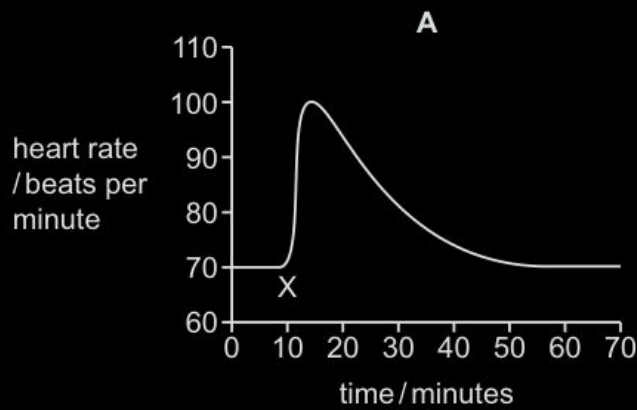
	focusing on a distant object	focusing on a near object
A	1, 3, 5, 8	2, 4, 6, 7
B	1, 4, 6, 8	2, 3, 5, 7
C	2, 3, 5, 7	1, 4, 6, 8
D	2, 4, 6, 7	1, 3, 5, 8

24	D	1
----	---	---

25 A student is walking along a road when a friend, who is hiding, jumps out suddenly giving her a shock.

Which graph shows the effect of adrenaline on the heart rate of the student as a result of the shock?

(X on the graph is the point at which the student's friend jumps out suddenly.)



MS 5090/12/M/J/24 Q25

25	A	1
----	---	---

5090/12/M/J/24 Q26

26 What is a sign of Type 1 diabetes?

- A lack of haemoglobin
- B raised blood glucose levels
- C reduced urine production
- D too much insulin

MS 5090/12/M/J/24 Q26

26	B	1
----	---	---

23 Nerve impulses in neurones can travel:

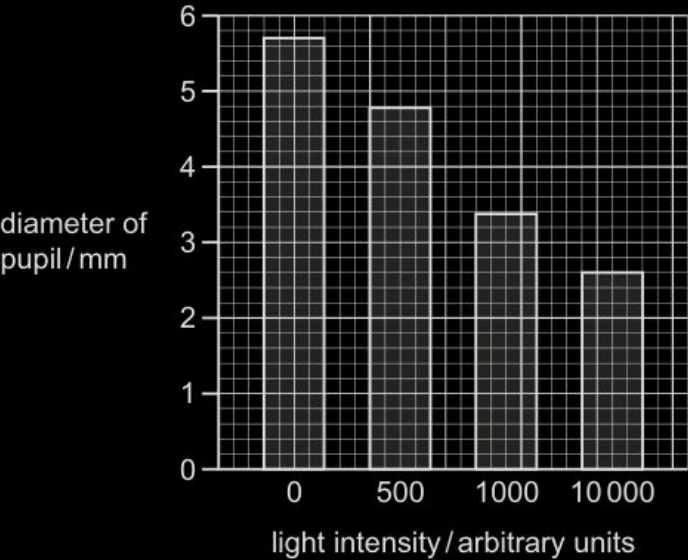
- 1 away from the central nervous system
- 2 towards the central nervous system
- 3 within the central nervous system.

In which direction do impulses in sensory and in relay neurones travel?

	sensory neurones	relay neurones
A	1	2
B	1	3
C	2	1
D	2	3

23	D	1
----	---	---

24 The diameter of the pupil of the eye was measured at a range of different light intensities.



What is the percentage decrease in pupil diameter (rounded to 1 decimal place) when a person moves from an area of complete darkness to an area of light intensity 10 000 arbitrary units?

- A 45.6% B 53.4% C 54.3% D 54.4%

24	D	1
----	---	---

25 When a person is frightened, which substance causes an increase in the blood sugar level?

- A adrenaline
- B carbon dioxide
- C insulin
- D lactic acid

MS 5090/11/O/N/24 Q25

25	A	1
----	---	---

5090/11/O/N/24 Q26

26 Raynaud's syndrome is a condition that can be triggered by stress. It causes the hands and feet to react as if they were affected by extreme cold.

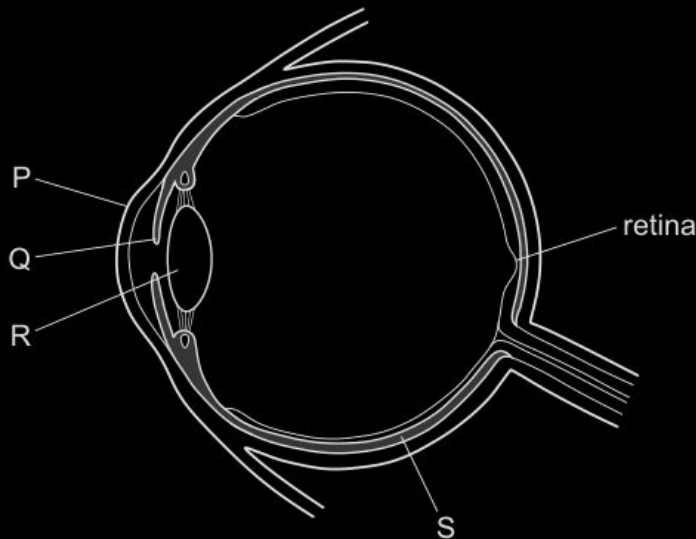
Which reaction will take place in the hands and feet of a person with Raynaud's syndrome?

- A vasoconstriction – blood vessels at the skin surface narrow, reducing blood flow
- B vasoconstriction – blood vessels at the skin surface widen, increasing blood flow
- C vasodilation – blood vessels at the skin surface narrow, reducing blood flow
- D vasodilation – blood vessels at the skin surface widen, increasing blood flow

MS 5090/11/O/N/24 Q26

26	A	1
----	---	---

24 The diagram shows a section through the eye.

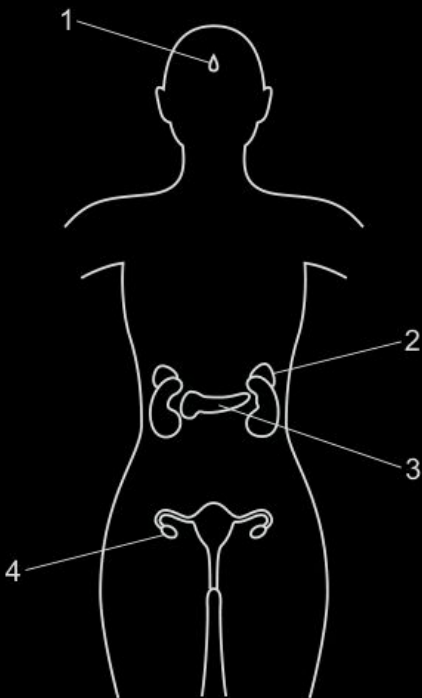


Which two structures focus light rays onto the retina?

- A** P and Q **B** P and R **C** Q and R **D** Q and S

24	B	1
----	---	---

25 The diagram shows some of the endocrine glands of a human female.



Which row matches the glands with the correct hormones?

	1	2	3	4
A	adrenaline	FSH	oestrogen	insulin
B	FSH	adrenaline	insulin	oestrogen
C	insulin	oestrogen	adrenaline	FSH
D	oestrogen	insulin	FSH	adrenaline

25	B	1
----	---	---

26 When the body temperature is too high, which process causes the body temperature to return to normal?

- A contraction of hair erector muscles
- B contraction of skeletal muscles, causing shivering
- C secretion of sweat onto the surface of the skin
- D vasoconstriction of arterioles supplying skin surface capillaries

26	C	1
----	---	---

25 Where are the receptors for the pupil reflex located?

- A brain
- B iris
- C pupil
- D retina

MS 5090/11/M/J/25 Q25

25	D	1
----	---	---

5090/11/M/J/25 Q27

27 Which statements about synapses are correct?

- 1 Synapses are junctions between neurones.
- 2 Synapses ensure impulses travel in only one direction.
- 3 Neurotransmitter chemicals diffuse across synaptic gaps.

- A 1, 2 and 3 B 1 and 2 only C 1 and 3 only D 2 and 3 only

MS 5090/11/M/J/25 Q27

27	A	1
----	---	---

28 Which process is **not** part of a homeostatic mechanism in mammals?

- A change in flow of blood in vessels of the skin
- B growth of longer hairs from the skin
- C increased rate of sweat production
- D contraction of hair erector muscles

MS 5090/11/M/J/25 Q28

28	B	1
----	---	---

Question	Answer	Marks
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5090/11/M/J/25 Q29

29 A nurse tests a sample of urine from a patient using Benedict’s reagent. The sample tests positive.

What is a possible explanation for this result?

- A The patient’s liver is **not** deaminating protein efficiently.
- B The patient’s pancreas is **not** producing sufficient insulin.
- C The patient has eaten too much glucose.
- D The patient’s concentration of plasma proteins is too high.

MS 5090/11/M/J/25 Q29

29	B	1
----	---	---

5090/11/M/J/25 Q37

37 Which row shows what happens in the eye when focusing on a nearby object?

	ciliary muscle	shape of lens
A	relaxes	becomes thinner
B	relaxes	becomes rounder
C	contracts	becomes thinner
D	contracts	becomes rounder

MS 5090/11/M/J/25 Q37

37	D	1
----	---	---

27 Which part of the eye contains the greatest density of light receptors?

- A cornea
- B optic nerve
- C iris
- D fovea

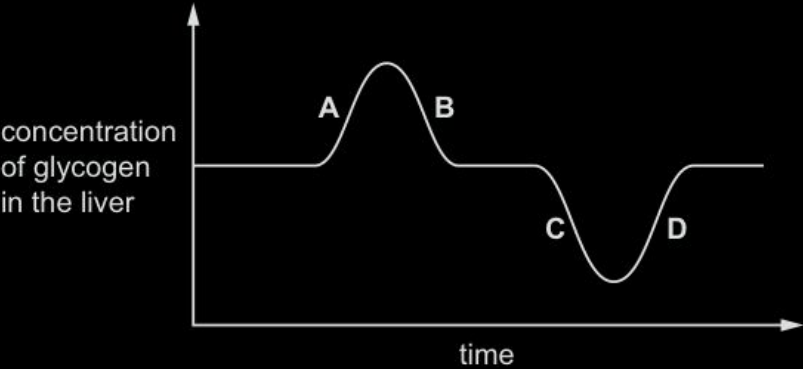
MS 5090/12/M/J/25 Q27

27	D	1
----	---	---

5090/12/M/J/25 Q30

30 The graph shows the concentration of glycogen stored in the liver of a human.

During which period of time is the secretion of adrenaline causing the blood glucose concentration to move towards its set point?



MS 5090/12/M/J/25 Q30

30	B	1
----	---	---

5090/12/M/J/25 Q31

31 The diagram shows one of the endocrine glands in the human body, labelled X.

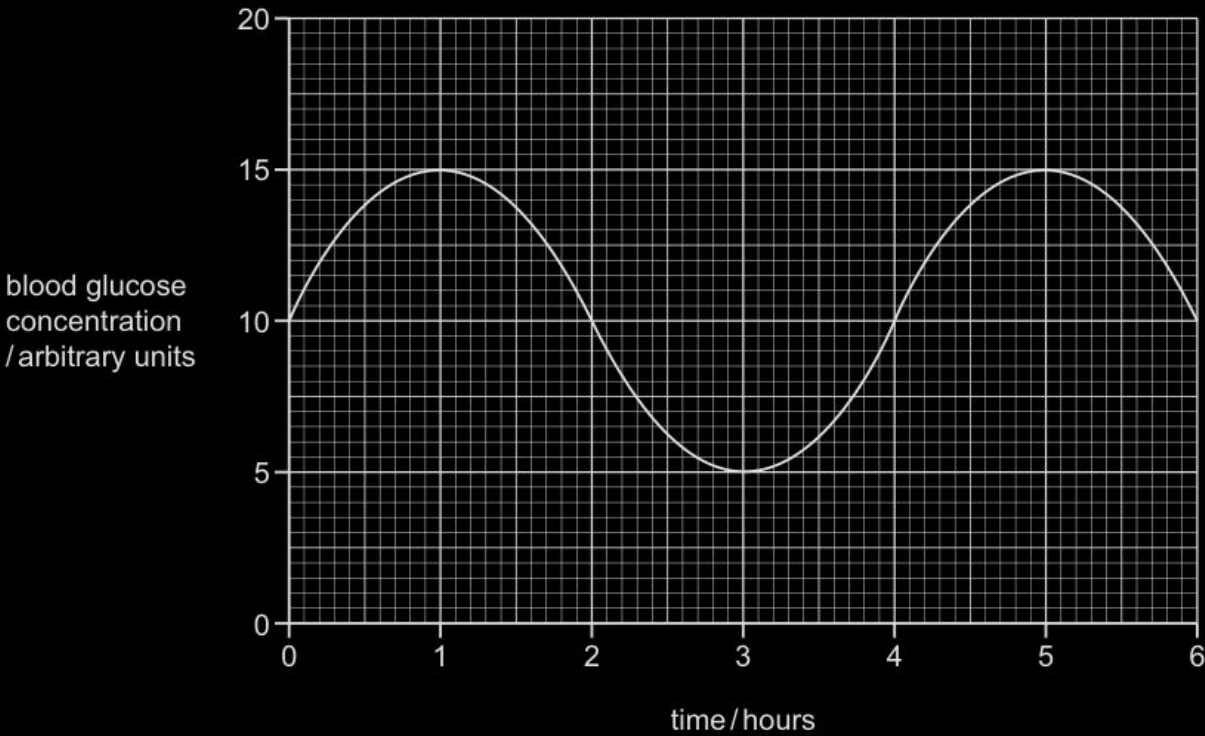


Which hormone is produced by endocrine gland X?

- A adrenaline
- B glucagon
- C luteinising hormone
- D progesterone

31	C	1
----	---	---

22 The graph shows blood glucose concentration over 6 hours in a day.



What is the set point for blood glucose concentration shown by this graph?

- A 5 arbitrary units
- B 10 arbitrary units
- C 15 arbitrary units
- D 20 arbitrary units

22	B	1
----	---	---

23 Which hormone can be used to treat Type 1 diabetes?

- A adrenaline
- B oestrogen
- C glucagon
- D insulin

23	D	1
----	---	---

24 An athlete’s resting body temperature is 37.0 °C.

After vigorous exercise for 30 minutes, the athlete’s body temperature rises to 38.6 °C.

The athlete rests for 10 minutes.

After resting, the athlete’s body temperature is measured again and is 37.2 °C.

Which events take place in the athlete’s body during the 10-minute rest to lower their body temperature?

	arterioles near the surface of the skin	heat energy is lost through
A	contract	evaporation of sweat
B	contract	shivering
C	dilate	evaporation of sweat
D	dilate	shivering

24	C	1
----	---	---

25 Which endocrine gland produces follicle-stimulating hormone?

- A adrenal gland
- B ovary
- C pancreas
- D pituitary gland

25	D	1
----	---	---

25 Which event happens when the blood glucose concentration rises?

- A Less adrenaline is released.
- B Less insulin is released.
- C More adrenaline is released.
- D More insulin is released.

25	D	1
----	---	---

26 What is **not** an example of homeostasis involving a negative feedback mechanism?

- A maintaining the core body temperature at around 36.5 °C
- B releasing adrenaline from the adrenal glands when frightened
- C releasing glucose from liver cells into the blood after missing a meal
- D releasing insulin from the pancreas after eating a meal

26	B	1
----	---	---